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Vu Cong Duy

Data Scientist/ AI Engineer

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Data Scientist with 4+ years of production experience in fintech and on-demand logistics. Specializes in causal inference, credit scoring, and end-to-end ML systems—from spatio-temporal feature engineering on telecom data to real-time pricing APIs. Key results: **+3.12% Gini** uplift on credit models and **40% GTV** increase in dynamic pricing A/B tests.

SKILLS

Programming languages	Skilled in **Python and SQL**, with a working knowledge of **Java and Scala**.
Tech stack	dbt, Vertex AI, BigQuery, PostgreSQL, Ray, Elasticsearch, Grafana, Spark, Airflow
Statistical	Real-world experience in Causal Inference, Time-series models, and Tree-based models.
Languages	English — **TOEIC L&R 895**.

EXPERIENCE

Data Nest **11/2025 – Present**

Data Scientist

- **Lead generation (lookalike model)** — find new prospects who resemble high-value customers
  - Defined the modeling target (strategic proxy labels) from the business process
  - Built the end-to-end training pipeline and a lead-quality evaluation metric
- **Location features for credit scoring** — **PySpark** pipeline turning telecom data into credit-risk signals
  - Converts ∼8.7B VNPT logs/week (∼16M subscribers) into mobility & location-diversity features (radius of gyration, spatial entropy) plus area-wealth enrichment
  - Feeds these new behavioral signals into loan-default models for VNPT credit clients
- **HDBank BNPL credit model** — **+1.76 Gini** (and **+2.17** on credit card)
  - Fixed a data leakage in the legacy evaluation (temporal train/test split)
  - Added a call-timing behavioral feature; used **PySpark ML** to select the top 50 of a 200+ feature inventory

Deliveree on-demand logistics **10/2024 – 11/2025**

Data Scientist

- **Dynamic Pricing (built from scratch)**: Achieved the highest recorded GTV increment of **40%** in A/B testing.
  - **Price–demand model (causal)**: From zero to production — set the goal, collected/cleaned event data, built a model to measure how price changes affect demand, ran experiments, and deployed with **Ray Job**. Added shadow rollout and monitoring.
  - **Potential demand model**: Changed the forecast horizon from 1h to 3h (with upsampling) based on uncertainty analysis, improving statistical soundness of pricing decisions.
  - Supply model: Designed a new supply model to replace the old "not-fulfilled" model. Proved via causal inference that price has no significant effect on the new target, relaxing the causality assumption on supply.
  - **Price safety**: Proposed and tested a statistical guardrail, running as a near-real-time **Ray Job** (15-min refresh), to detect market reactions and cap prices before the system diverges in edge cases.
  - **Competitive pricing**: Helped design a separate near-real-time **Ray Job** (15-min refresh) that integrates crawled competitor prices into the pricing logic, keeping our prices competitive against the market.
  - **Engineering**: Built the pricing API with **Ray Serve**, trained models on **Google Vertex AI Pipelines**, set up **Grafana** dashboards for monitoring, and designed the model output pipeline on PostgreSQL to ensure data availability. Partnered with DevOps on the model-deployment CI/CD, proposing two serving strategies and weighing the trade-off between deploy flexibility (load model from URI at runtime) and immutability/startup reliability (bundle the model into the image).
- **Driver matching system (enhancement)**: Audited driver portfolio data and ran cohort-level error analysis to identify underperforming segments. Refactored the sampling pipeline, cutting training time from **12 hours to under 2 hours**.

Timo Digital Bank **04/2023 – 06/2024**

Data Scientist

- Developed an expense prediction model by redefining the target variable and implementing a data pipeline with **dbt** and **Airflow**. Delivered insights through dashboards on **Holistics** and personalized customer engagement using **Mandrill**.
- Enhanced transaction purpose classification by applying **rule mining**, **Elasticsearch** (TF-IDF, Bayesian clustering), and advanced evaluation with **EvidentlyAI**. Deployed the improved model as an API service using **SpringData**.
- Designed and implemented graph algorithms (**DFS, BFS, heuristic growth**) on 4TB of user interaction data with **PySpark**, uncovering customer behavior patterns that informed UI/UX improvements in money transfer and contact-adding flows.

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Thesis: Speech Enhancement using Deep Learning on Edge Devices **01/2023 – 06/2023**

- Applied DCCRN model with feature fusion and similarity-preserved knowledge distillation to compress a speech enhancement model for edge deployment. Implemented complex-number neural networks in PyTorch and optimized multi-GPU training with PyTorch Lightning. **Grade: 9.1/10** [thesis]

Vietnam International Bank (VIB) — Customer Data Platform **09/2022 – 01/2023**
Vendor Data Scientist *Bluebolt Software*

- Scoped DS use cases (churn prediction, spending regression, recommendation) by auditing data across VIB's card, digital, and marketing departments. Assessed data quality and built SAP dashboards to track DS metrics for the BA team.

EDUCATION

Bachelor of Computer Science, Specialized in Data Science **04/2019 – 08/2023**
HCM University of Science - VNU, GPA: **8.46/10**